BADO LEE (이바도)

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SUMMARY

AI researcher and engineering leader specializing in vision-language and omnimodal foundation models, with 15+ years of experience across Samsung Electronics, NAVER, and NAVER Cloud. Currently Head of the Multimodal Model organization at NAVER Cloud building the HyperCLOVA X SEED foundation-model family. Co-creator of CRAFT, a widely used open-source scene-text detector (CVPR 2019; 1,671 citations; default detection engine of EasyOCR). 2,805 total citations (h-index 12) and 7 registered patents.

EDUCATION

Seoul National University — M.S. in Electrical and Computer Engineering Mar 2010 – Feb 2012

- Biointelligence Laboratory (Advisor: Prof. Byoung-Tak Zhang); hypernetwork-based multimodal concept learning.

Seoul National University — B.S. in Electrical Engineering (Electrical and Information Engineering) Mar 2003 – Feb 2010

PROFESSIONAL EXPERIENCE

NAVER Cloud Corp. Seongnam, Republic of Korea | Jan 2023 – Present

Head, Multimodal Model Organization Jan 2026 – Present

- Lead the Multimodal Model team, whose groups cover foundation-model pre-training, cross-modal alignment, generation, and data, with final responsibility for architecture, training strategy, and evaluation of the HyperCLOVA X SEED vision-language and omnimodal foundation models.
- Released HyperCLOVAX-SEED-Omni-8B as open source on Hugging Face (Jan 2026).
- Lead the multimodal workstream of NAVER's participation in the NVIDIA Nemotron Coalition — the first Korean company among 12 global AI companies (incl. Cursor, Mistral AI, Perplexity) co-developing open models across pre-training, post-training, and reinforcement learning.

VLM Development Lead — Sovereign AI Foundation Model Project (Ministry of Science and ICT) Jun 2025 – Dec 2025

- Led vision-language modality development in Korea's national sovereign-AI foundation model program, a government-run competition among five national consortia.

Lead, Vision-Language Models (Semantic Perception) Apr 2023 – May 2025

- Built the full VLM pipeline for the HyperCLOVA X family: vision encoder, multimodal pre-training (MMPT), and SFT/RL post-training.
- Released HyperCLOVAX-SEED-Vision-Instruct-3B as open source (~120K monthly downloads on Hugging Face as of Aug 2026).

Lead, OCR Jan 2023 – Mar 2023

- Continued leading the OCR organization after its transfer from NAVER Corp. to NAVER Cloud.

NAVER Corp. Seongnam, Republic of Korea | Jan 2018 – Dec 2022

Lead, OCR / Document AI Sep 2020 – Dec 2022

- Led document-AI engine development — CORD (receipt-parsing dataset), DEER (detection-agnostic end-to-end recognizer), CREPE (coordinate-aware document parser) — deployed in the commercial CLOVA OCR service.
- Commercial impact: automated medical-expense receipt processing for Heungkuk Fire & Marine Insurance; won the National Diet Library of Japan digitization project against incumbent Japanese OCR vendors.

Software Engineer, OCR — Clova AI Jan 2018 – Aug 2020

- Co-created the CRAFT scene-text detector (CVPR 2019) and commercialized it in CLOVA OCR.
- Developed few-shot handwriting font-generation technology; launched the "Nanum Handwriting" campaign (2019): ~25,000 public participants and 109 AI-generated fonts released free of charge.

Samsung Electronics Co., Ltd. — Mobile Communications Business Aug 2012 – Dec 2017

Engineer, Vision Engine

- Developed the on-device computer-vision engine (image recognition and processing) for Galaxy smartphone cameras; shipped in commercial devices.

Seoul National University — Biointelligence Laboratory Mar 2010 – Feb 2012

Graduate Researcher (government-funded Doyak Research Program, Ministry of Education, Science and Technology)

- Researched hypernetwork-based multimodal concept learning — cross-modal text–image generation and hierarchical latent-variable models for video story analysis; 8 publications.

AWARDS — N Innovation (NAVER company-wide technology competition)

- 2022 — DEER: Top 3, Service Development track
- 2021 — Document Recognition: selected as Outstanding Technology
- 2020 — Receipt CAPTCHA: Service Development track
- 2019 — Handwriting Recognition: selected as Outstanding Technology
- 2018 — OCR: competition award

OPEN-SOURCE & INDUSTRY IMPACT

- CRAFT-pytorch (github.com/clovaai/CRAFT-pytorch, MIT license): ~3.4K GitHub stars and ~930 forks (Aug 2026); adopted as the default text-detection model of EasyOCR (~29.8K stars), a widely used open-source OCR library.
- HyperCLOVA X SEED multimodal models on Hugging Face (huggingface.co/naver-hyperclovax): Vision-Instruct-3B — ~120K monthly downloads, 221 likes; Omni-8B — 192 likes (Aug 2026).
- CLOVA OCR in production across NAVER services: MyPlace receipt reviews, NAVER WORKS, Remember (business-card app), CLOVA Lamp, Smart Place, and VIBE.
- Nanum Handwriting (2019): 25,000 public participants; 109 free fonts in nationwide use.

PUBLICATIONS

Google Scholar (Aug 2026): 28 papers · 2,805 citations · h-index 12 · i10-index 12.

International Conferences & Journals

1. Y. Baek, **B. Lee**, D. Han, S. Yun, H. Lee. "Character Region Awareness for Text Detection (CRAFT)." IEEE/CVF CVPR 2019, pp. 9365–9374. — 1,671 citations

2. S. Park, S. Shin, **B. Lee**, J. Lee, J. Surh, M. Seo, H. Lee. "CORD: A Consolidated Receipt Dataset for Post-OCR Parsing." NeurIPS 2019 Workshop on Document Intelligence. — 423 citations

3. S. Park, S. Chun, J. Cha, **B. Lee**, H. Shim. "Multiple Heads are Better than One: Few-shot Font Generation with Multiple Localized Experts (MX-Font)." IEEE/CVF ICCV 2021, pp. 13880–13889. — 181 citations

4. S. Park, S. Chun, J. Cha, **B. Lee**, H. Shim. "Few-shot Font Generation with Localized Style Representations and Factorization (LF-Font)." AAAI 2021, 35(3), pp. 2393–2402. — 175 citations

5. J. Cha, S. Chun, G. Lee, **B. Lee**, S. Kim, H. Lee. "Few-shot Compositional Font Generation with Dual Memory (DM-Font)." ECCV 2020, pp. 735–751. — 159 citations

6. W. Hwang, et al. (incl. **B. Lee**). "Post-OCR Parsing: Building Simple and Robust Parser via BIO Tagging." NeurIPS 2019 Workshop on Document Intelligence. — 60 citations

7. S. Park, S. Chun, J. Cha, **B. Lee**, H. Shim. "Few-shot Font Generation with Weakly Supervised Localized Representations." IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) 46(3), pp. 1479–1495. — 43 citations

8. J.-W. Ha, B.-H. Kim, **B. Lee**, B.-T. Zhang. "Layered Hypernetwork Models for Cross-modal Associative Text and Image Keyword Generation in Multimodal Information Retrieval." PRICAI 2010, pp. 76–87. — 21 citations

9. S.-J. Kim, J.-W. Ha, **B. Lee**, B.-T. Zhang. "Evolutionary Layered Hypernetworks for Identifying microRNA–mRNA Regulatory Modules." IEEE Congress on Evolutionary Computation (CEC) 2010, pp. 1–8. — 15 citations

10. J.-H. Lee, **B. Lee**, J.-S. Kim, R. Deaton, B.-T. Zhang. "A Molecular Evolutionary Algorithm for Learning Hypernetworks on Simulated DNA Computers." IEEE CEC 2011, pp. 2735–2742. — 14 citations

11. G. Kim, et al. (incl. **B. Lee**), S. Park. "CREAM: Visually-Situated Natural Language Understanding with Contrastive Reading Model and Frozen Large Language Models." EMNLP 2023. — 13 citations

12. S. Kim, et al. (incl. **B. Lee**), Y. Baek. "DEER: Detection-agnostic End-to-End Recognizer for Scene Text Spotting." arXiv:2203.05122, 2022. — 13 citations

13. Y. Okamoto, et al., S. Park, **B. Lee**. "CREPE: Coordinate-Aware End-to-End Document Parser." ICDAR 2024, pp. 3–20. — 7 citations

14. J. Cha, S. Chun, G. Lee, **B. Lee**, S. Kim, H. Lee. "Toward High-quality Few-shot Font Generation with Dual Memory." CVPR 2020 Workshop on AI for Content Creation. — 5 citations

15. **B. Lee**, H.-S. Seok, B.-T. Zhang. "Evolving a Population Code for Multimodal Concept Learning." IEEE CEC 2011, pp. 699–706.

16. **B. Lee**, H.-S. Seok, B.-T. Zhang. "Video Streams Semantic Segmentation Utilizing Multiple Channels with Different Time Granularity."

Domestic (Korean) Journals & Conferences

17. J.-W. Ha, H. Park, **B. Lee**, M. Hwang. "NAVER AI Platform CLOVA and Hyperscale AI HyperCLOVA." Communications of KIPS 28(3), pp. 56–66, 2021 (in Korean).

18. J.-W. Ha, B.-H. Kim, **B. Lee**, B.-T. Zhang. "Auto-tagging Method for Unlabeled Item Images with Hypernetworks for Article-related Item Recommender Systems." Journal of KIISE: Computing Practices and Letters 16(10), pp. 1010–1014, 2010. — 2 citations

19. B.-H. Kim, **B. Lee**, S. Ha, N.-I. Cho, B.-T. Zhang. "Article–Product Associated Search Using a Bi-Source Topic Model." Journal of KIISE: Computing Practices and Letters 17(4), pp. 249–253, 2011 (in Korean). — 1 citation

20. **B. Lee**, H.-S. Seok, B.-T. Zhang. "Hierarchical Latent-Variable Models for Story Analysis of Drama Videos." Journal of KIISE: Computing Practices and Letters 18(5), pp. 434–438, 2012 (in Korean).

21. H.-S. Seok, **B. Lee**, B.-T. Zhang. "A Nonparametric Bayesian Method for Multi-channel-based Semantic Segmentation of Drama Videos." Proc. KIISE Conference 39(1B), pp. 474–476, 2012 (in Korean).

22. **B. Lee**, H.-S. Seok, B.-T. Zhang. "Hierarchical Latent-Variable Models for Story Analysis of Drama Videos." Proc. KIISE Conference 38(2B), pp. 267–270, 2011 (in Korean).

23. **B. Lee**, B.-T. Zhang. "Efficient Image Representation Using Topic Modeling." Proc. KIISE Conference 38(1C), pp. 319–322, 2011 (in Korean).

24. **B. Lee**, H.-S. Seok, B.-T. Zhang. "Sentence-Generation Capability of Random-Hypergraph-based Language Models Considering Inter-word Distance Relations." Proc. KIISE Conference, pp. 278–281, 2010 (in Korean).

25. J.-W. Ha, B.-H. Kim, **B. Lee**, B.-T. Zhang. "Auto-tagging Method for Unlabeled Product Images with Hypernetworks for Article-related Product Recommender System." Proc. KIISE Conference, pp. 123–124, 2010.

PATENTS (7 registered — 5 Korea, 2 Japan; assigned to NAVER / NAVER Cloud)

1. Method and System for Providing a Handwriting Font Generation Service — KR 10-2501631; JP 7348446
2. Method and System for Font Generation — KR 10-2376231; JP 7250734
3. Method and System for Providing CAPTCHA — KR 10-2313162
4. Image Distortion Correction Method and Object Collection Method Using the Same — KR 10-2279259
5. Apparatus, Method, and System for Character Detection — KR 10-2211763

INVITED TALKS

- DEVIEW 2019 (Oct 28, 2019, COEX, Seoul) — "Building an AI That Writes My Handwriting for Me." DEVIEW is Korea's largest developer conference, hosted by NAVER. (deview.kr/2019/schedule/294)
- DEVIEW 2020 — "CAPTCHA vs. PARSING." (deview.kr/2020/sessions/406)

MEDIA COVERAGE

- Digital Daily (Feb 3, 2022). "[NAVER Avengers] Receipt Reviews and Business-Card Apps — You Are Already Using CLOVA OCR." In-depth interview series on NAVER's technology leaders; featured as one of three interviewees (OCR Team Lead). ddaily.co.kr/page/view/2022020316413084008
- ZDNet Korea (Oct 28, 2019). "200 Handwritten Characters Become a Font — NAVER's Handwriting Technology Reaches Commercialization." Solo feature with interview. zdnet.co.kr/view/?no=20191028132832
- The Electronics Times / ETNews (Jun 21, 2019). "NAVER Proves World-Class AI Capability Again at CVPR." Named contributor for CRAFT. etnews.com/20190621000179
- Segye Ilbo (Jun 23, 2019). "NAVER Participates in CVPR 2019, the Most Prestigious Venue in Vision and Deep Learning." Named contributor. segye.com/newsView/20190621504813

- AI Times (Oct 18, 2021). "ICCV 2021 — NAVER Presents a Record 13 Papers, Proving AI Leadership." Named contributor for MX-Font. aitimes.kr/news/articleView.html?idxno=22763
- Economic Review (Apr 27, 2020). "[IT Curation] OCR, and NAVER." Quotes my DEVIEW talk. econovill.com/news/articleView.html?idxno=394610
- SBS Biz (Dec 30, 2025). "Who Earns the National Emblem — Vetting Begins for Korea's 'National AI'." Broadcast coverage of the sovereign AI foundation-model program. biz.sbs.co.kr/article/20000281693
- Hankyoreh 21 (Oct 9, 2020). "[Hangul Day] Write Just 28 Characters and Get a Handwriting Font in an Hour." h21.hani.co.kr/arti/culture/culture_general/49322.html